Group 11: Healthcare Industry

Tobacco Use and Health Risks: Leveraging Big Data to Analyze Behavioral Patterns and Trends

Project contribution

1. Nandini Nagiri

In this project I was mostly involved in data collection whereby data was obtained from the BRFSS Survey via the CDC's STATE System. I was responsible for acquiring and incorporating both archival data about smoking history and real-time data streaming, thus, the project includes extensive patterns of tobacco usage up to date.

To this end, I created a secure data pipeline that allowed seamless data ingest into Google Cloud Storage. This particular pipeline was instrumental in transitioning from data integration and prepared this data for the entire team in a simple but secure fashion. I made certain that this storage solution is set up for its safety and also its efficiency where authorization permissions, and the other pre-requisites for the authorized were concerned.

But not only the data themselves had to be preserved; I also optimized the form of the dataset for analysis purposes. This kind of structure made it flexible enough so that the team could run queries and gain insights into the demographic trends. Structured data is the important work that laid the groundwork for analysis and allowed the team to produce valuable information on tobacco use in different population subgroups.

In this context, I was able to help the project to process thousands of records in a secure manner while providing context for the team to pursue high-volume research with accuracy. Some of the measures that I took to achieve them included facilitating easy access to the data and at the same time guaranteeing triennial validity in terms of demographic characteristics of tobacco use. In general, my work set the stage for the project's data analysis and involved at least some measure of generating insights to help identify at-risk groups and temporal tendencies in tobacco usage.

2. Vamshi Kalyan Yerramilli

The key activities that I have performed for this project were data cleaning and data preprocessing. While using PySpark, I dealt with data quality issues, dealt with missing values and preprocessed the given data set for later extraction in BigQuery and Hive. The raw data was made ready for analysis work through me; this involves identifying problems or errors that directly influence the result.

However, apart from cleaning, I concentrated on creating fields mainly concerning the prevalence of tobacco usage, the frequency, and mix 2 quit attempts. These fields were grouped under demography; this gave the team an easily understandable demographic facet of behaviour in the use of tobacco. This involved preparing the data and putting it into a format that would be analysed to provide finer features around the patterns of tobacco use by different demographic variables;

age, gender and education level. Some of the preprocessing I did involved differences in data format where I was needed to standardize data that was in different formats to ensure that all data fields in the data sets matched the required analysis format I had to deal with this. I also consulted the team on the dataset for the project and ensuring that the dataset was satisfactory on the standards of accuracy and relevance. Therefore, concentrating on reliable and quality data, I contributed to the generation of a reliable dataset for deeper analysis of tobacco consumption patterns.

In view of these, I was able to enhance the dataset to offer substantive groundwork for analyzing tobacco use by different people accurately and with more precision. My work was instrumental in facilitating arriving at the insights that would inform public health interventions of various at-risk populations. This process provided the foundation for valid and reliable conclusions based on data and gave the team the ability to identify demographic specific usage patterns of tobacco and help in developing corresponding strategies for interventions.

3. Vijay Venkatarao Goparaju

During this project, I had to make decisions on how to organize data processing environment for our project while establishing environment for large datasets. To achieve this, I adjusted Hadoop, Hive and Spark on Dataproc cluster of Google Cloud where our team could operate with both static and streaming data of BRFSS. My objective was to design a system that would be able to support tasks that involve analysis by getting through lots of data at a single instance.

To achieve this I reduced dependencies and made the setup highly scalable so that the infrastructure could handle data queries and complex analysis. To enhance speed and efficiency of processing big and distributed data sets, I set Dataproc to process all tasks in parallel. Such arrangement was necessary for maintaining the data processing integrity and let our team perform the queries on very large datasets continuously.

To optimize it on a further level, I adjusted the configuration of Spark and Hive that affected the allocation of resources nodes. This way there were no interferences during the processing and the functions of the data ingestion, transformation and analysis were seamless. Through the use of the Spark History Server, which presents data about the jobs at the web page designed for the users, I could easily monitor the jobs, scrutinize defects in order to correct them and consequently monitor the resource utilization and the performances of the jobs.

Furthermore, I also ensured that they remained modular to aid in the organization of the BRFSS dataset's static and active real-time components. My work ensured that the data team's queries were processed in a fairly strong and stable computational platform to enable them derive the insights they need from queries within the shortest time possible. In light of this, my inputs put in place a high performance that was instrumental in the analytical complexities of the project particularly providing important insight into the data to be fed into the public health programs of the country.

4. Bala Phaneendra Pothuri

For this particular task, I had used some exploratory data analysis (EDA) to determine pre-transformations and significant trends of tobacco consumption in the specified demographical segments. Over several years I analyzed tobacco usage through BigQuery and Hive considering gender, education level, race and age. My main objective was to use easy to understand data

presentation to reveal information necessary for the development of public health policies in relation to the identified vulnerable population and levels of consumption.

From the trend analysis that I conducted, especial years like 2013 had extremely high tobacco consumption figures. This insight guided the team to concerns that maybe causing these spikes and explore them for public health implications. I further expanded my work to analyzing seasonal trends of tobacco use and was able to present data as to how this type of substance use changes over the course of the year and what extent may need prevention efforts.

In order to comprehend demographic characteristics that might be at higher risk I went through gender as well as education level usage profiles of tobacco. This analysis provided evidence that adult tobacco user rates were steadily higher in some categories more than others, based on principles such as gender and level of education. That way, such findings enabled the team to discuss specific preventive measures that would address the special vulnerability of the said employees.

My professional activities also involved presentation of such trends in a coherent manner because the essence of the data needed to be explained to the team and other interested parties. It also provided a perfect depiction of the behavioural pattern in use of tobacco products that could be used to easily draw recommendations from the data insights.

Thus, the results of my EDA served as the foundation for suggesting directed interventions in community health that would help minimize tobacco consumption among high-risk groups. Thus, as a result of analyzing specific demographics, special years, and seasons, I contributed specific data-driven recommendations that should be employed in future public health initiatives. In the end, its findings helped to promote the development and improvement of further methods treating tobacco as a hazard to public health.

5. Jaya Sai Krishna Paleru

Other tasks I had included tobacco risk analysis in view of the impact on the general population with special emphasis on different categories likely to be in the high risk category. Thus, while considering demographic attributes as a major factor of focus in my analysis, I was able to determine how the usage of tobacco differs across age strata, gender and educational background. My goals were to bring attention to important patterns that might help specifically guide interventions in public health.

Using critical evaluation, I realized that some groups, including male and people of some education level, used tobacco intensively. These results offered insights into those population subgroups at greater risk of negative health outcomes associated with tobacco use. To my knowledge, my work showed that the target high-risk drinkers require appropriate & relevant prevention interventions targeting behaviours & difficulties they encounter.

To these ends, I analyzed how these demographic patterns could be harnessed to inform the development of such programs that would effectively predispose the control of tobacco proliferation. Being able to identify these populations of continued elevated tobacco use, I provided information that could be used to construct cognate cessation and preventional programs. The purpose of this analysis was not only to look for the vulnerable population and quantify the

threat to the consumer population but also to examine the particular behavioral consistency that continues tobacco use.

Also, I evaluated how these demographic trends might affect the disease risks over the life years revealing the significance of the primary prevention. The informations I gathered helped the team to provide recommendations of concrete public health activities targeted at decreasing tobacco consumption and decreasing people's health risks.

All in all, my contributions were crucial for enhancing a strategic development of the effective intervention policies to respond to the current public health issues related to tobacco consumption. That way I was able to make sure that the public health actions towards reducing the of incidences of tobacco use and the health risks inherent with the substance, are appropriately targeted to recommended demographics.

Executive summary

In this project, "Understanding Tobacco Related Health Risks Through Behavioral Data Analysis", we look at the Health Risks associated with tobacco use by analyzing the Behavioral Risk Factor Surveillance System (BRFSS) Survey data from 2011 to 2019. The goal is to better understand the behavioral health consequences of tobacco use and find out the demographic characteristics most impacted by health conditions linked to tobacco use. The project takes a robust data analysis approach to produce actionable insights that guide public health strategies and interventions.

To understand tobacco consumption patterns completely, we used both static and streaming data in the data acquisition phase. Data used in the project are BRFSS Survey Data extracted from the CDC's State Tobacco Activities Tracking and Evaluation (STATE) System. The demographic factors relevant to prevalence, usage frequency and quit attempts for both cigarettes and e-cigarettes are included in this dataset. The CDC's STATE System provides for this kind of access to this data which, along with it, proved vital for the analysis of behavioral trends in tobacco usage.

The dataset was cleaned and transformed although it had to go through rigorous cleaning and transformation phases, and had to handle missing values, formatting and structuring the data, to ensure data quality and prepare for analysis. Queries were executed, insights were extracted, and many twists and turns were exposed in how tobacco is consumed in BigQuery and Hive. Key findings of this analysis include the disproportionate relative impact of tobacco use at any education level, and greater gender differences in usage. Furthermore, there are specific demographic groups (defined by race and age) which manifest unique patterns and trends during the period studied. These findings indicate that male and specific educational demography participation in tobacco use is high and makes it important to focus on public health campaigns. Furthermore, the analysis identified temporal trends, showing it was more likely that in 2013, the average consumption rate was higher. Trends were also contextualized within seasonal variations and used to help direct the development of targeted public health interventions.

Finally, this project highlights the need for further research and monitoring, for at risk demographic groups. One of the most important things we must do is provide targeted interventions for all these vulnerable populations. In addition to highlighting the continuing magnitude of tobacco's public health burden, this project lays the groundwork to develop intervention measures to reduce tobacco prevalence and mitigate health risks of its use. These insights can serve as important advice for

public health officers and lawmakers who are working to construct healthier communities free from tobacco related diseases.

Data acquisition

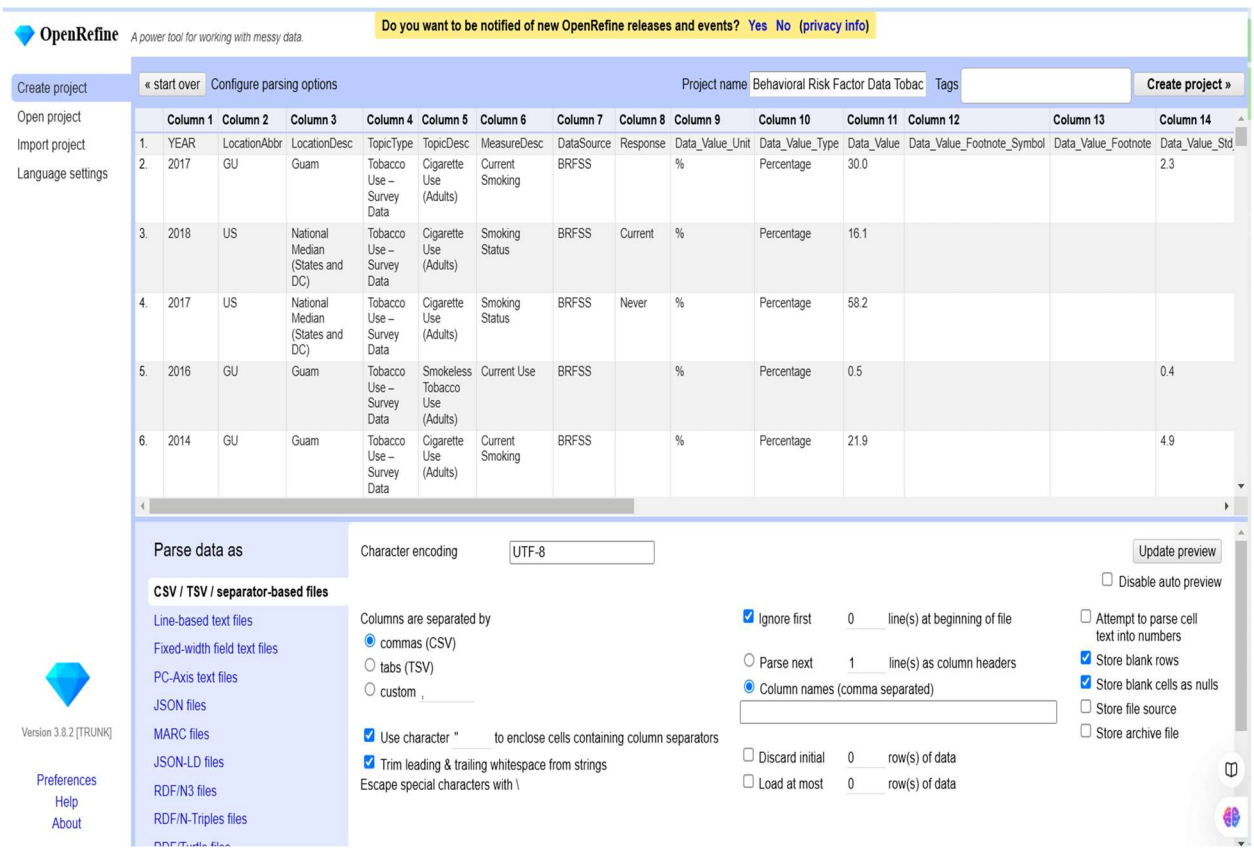
The project used BRFSS Survey Data as static data extracted from the CDC · State Tobacco Activities Tracking and Evaluation (STATE) System.

The data set used included only prevalence, use frequency and quit attempts of cigarettes and e-cigarettes according to demographics. Such data was vital towards understanding the behavioral pattern concerning the use of tobacco. The dataset can be accessed at the following link: [BRFSS Survey Data.]

For streaming data, we extracted data for further real-time analysis from this link. [Tobacco usage streaming data].

Before ingesting data into GCP . The first data transformation and formatting were done using open refine.

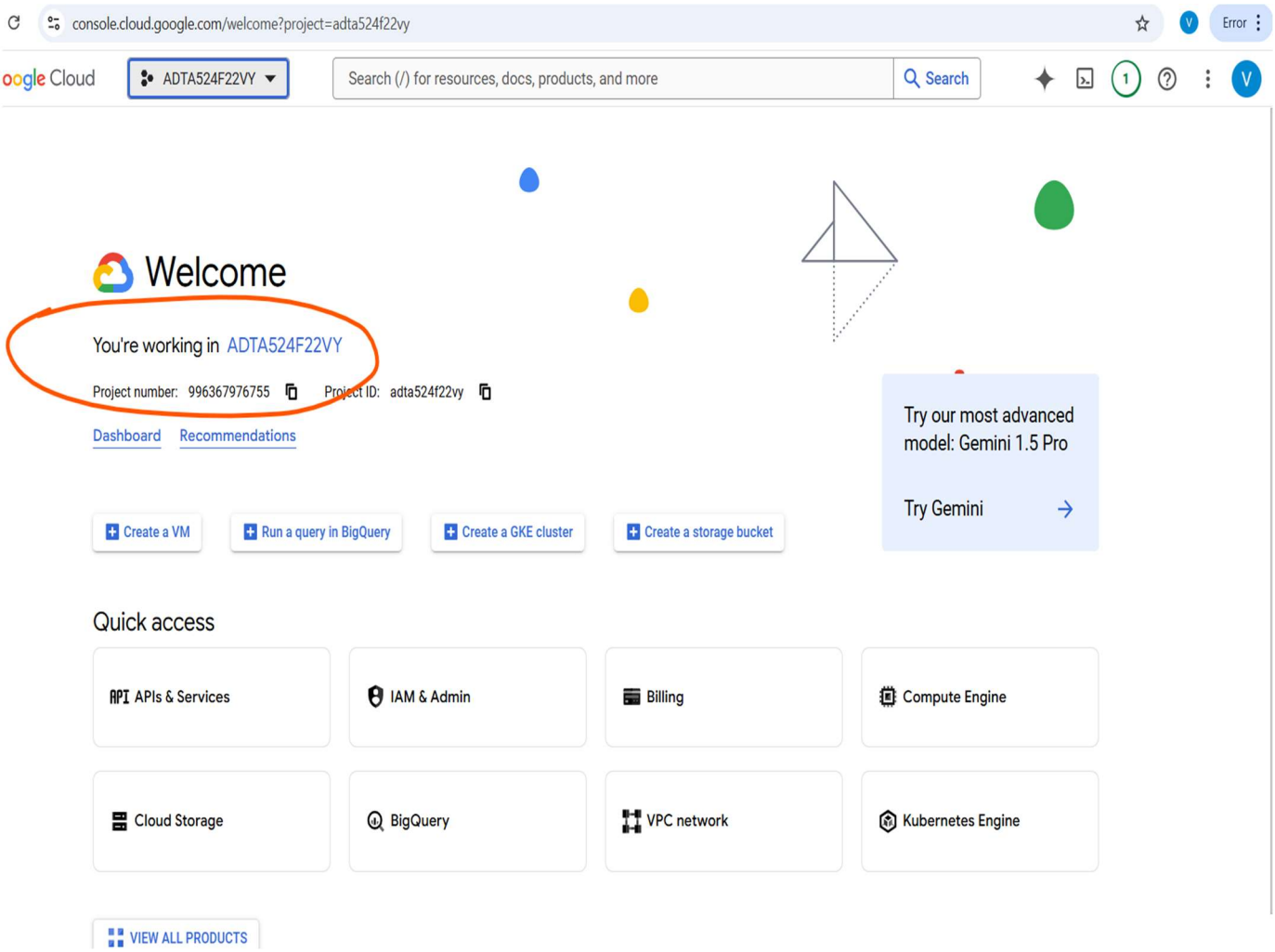
We trimmed all the white spaces and separated the columns with commas.



	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13	Column 14
1.	YEAR	LocationAbbr	LocationDesc	TopicType	TopicDesc	MeasureDesc	DataSource	Response	Data_Value_Unit	Data_Value_Type	Data_Value	Data_Value_Footnote_Symbol	Data_Value_Footnote	Data_Value_Std
2.	2017	GU	Guam	Tobacco Use – Survey Data	Cigarette Use (Adults)	Current Smoking	BRFSS		%	Percentage	30.0			2.3
3.	2018	US	National Median (States and DC)	Tobacco Use – Survey Data	Cigarette Use (Adults)	Smoking Status	BRFSS	Current	%	Percentage	16.1			
4.	2017	US	National Median (States and DC)	Tobacco Use – Survey Data	Cigarette Use (Adults)	Smoking Status	BRFSS	Never	%	Percentage	58.2			
5.	2016	GU	Guam	Tobacco Use – Survey Data	Smokeless Tobacco Use (Adults)	Current Use	BRFSS		%	Percentage	0.5			0.4
6.	2014	GU	Guam	Tobacco Use – Survey Data	Cigarette Use (Adults)	Current Smoking	BRFSS		%	Percentage	21.9			4.9

Step 1:setting up the GCP project environment

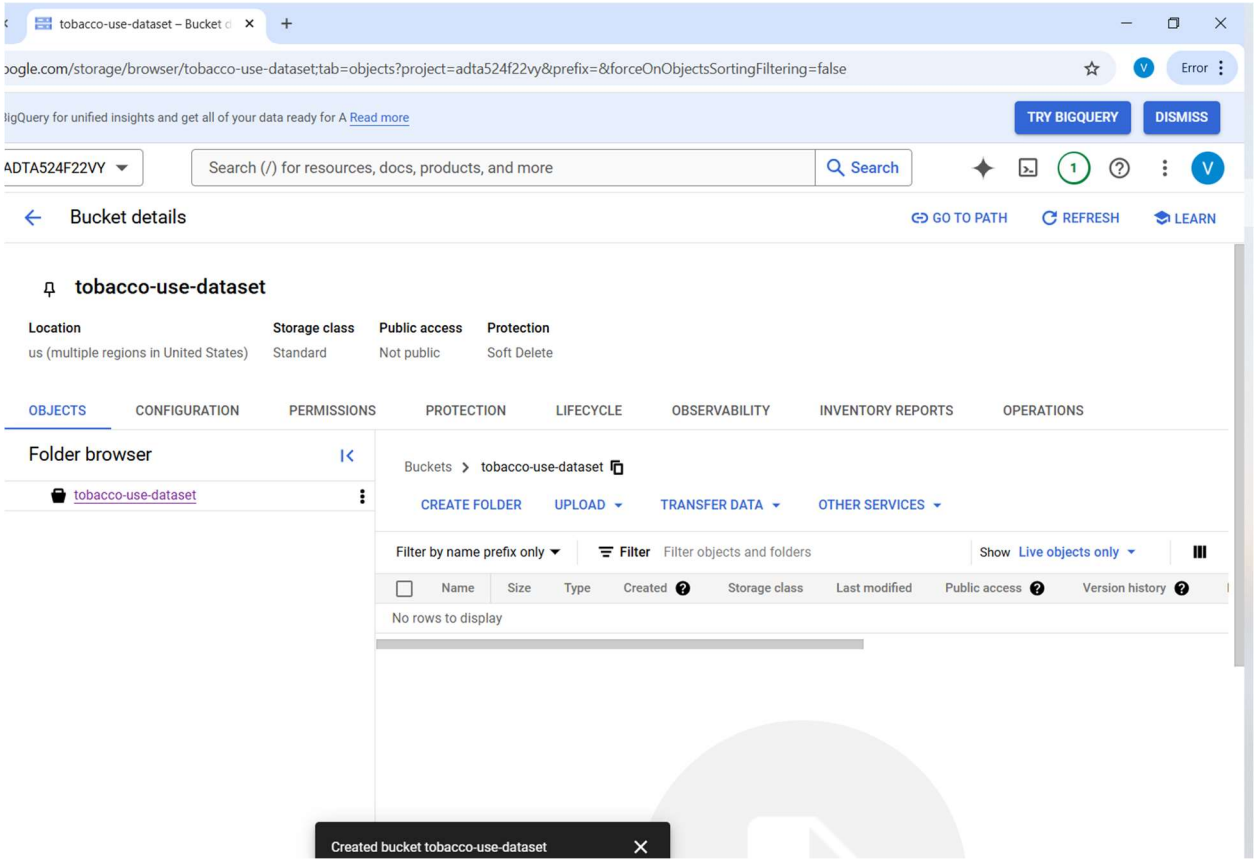
The first thing to do is to setup a GCP project which will hold all of the tasks and analyses needed to be managed efficiently from the start of the project till the end of the project lifecycle.



Step 2:Configuring the google cloud storage

Setting up the storage bucket for efficient data storage.

We created the bucket named **“tobacco-use-dataset”.**

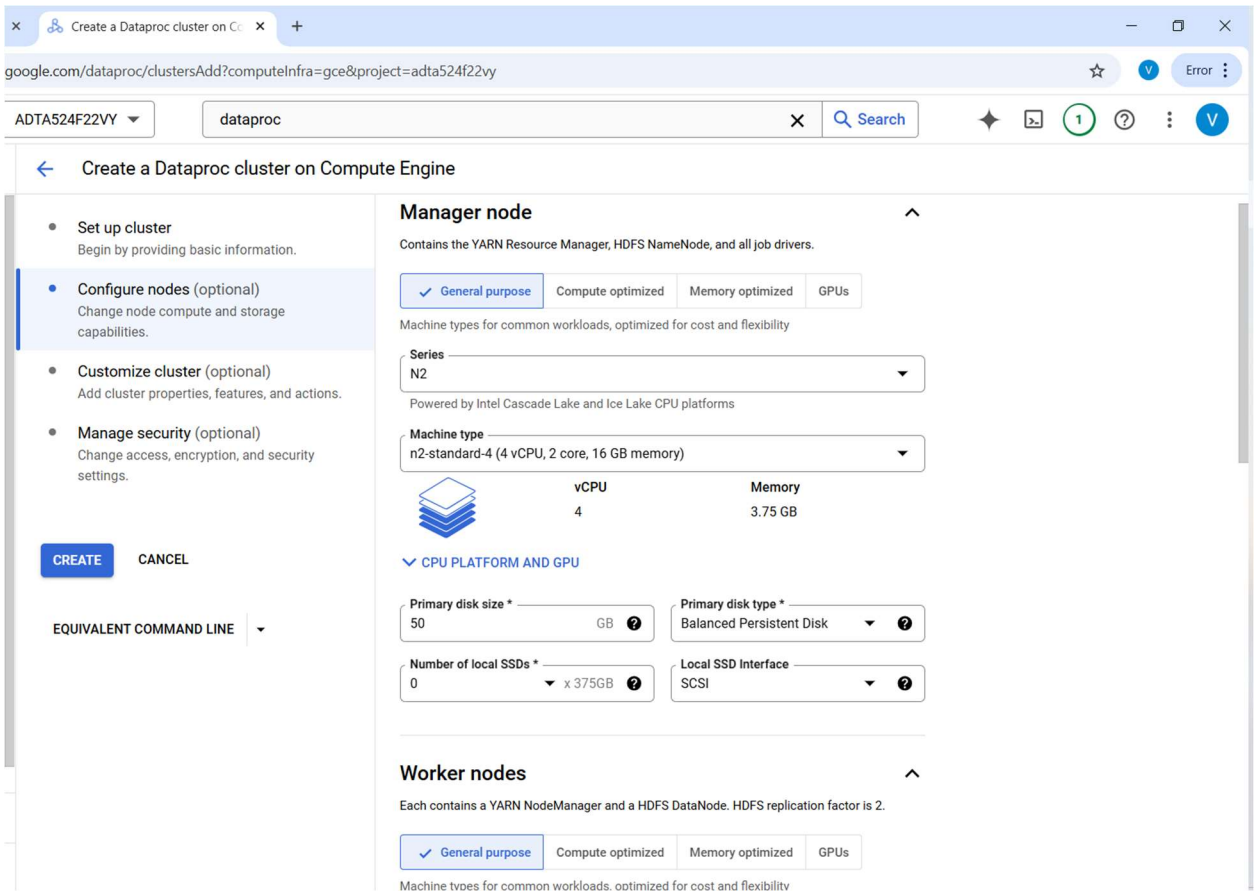


This also involved setting up the appropriate region and permission for team access.

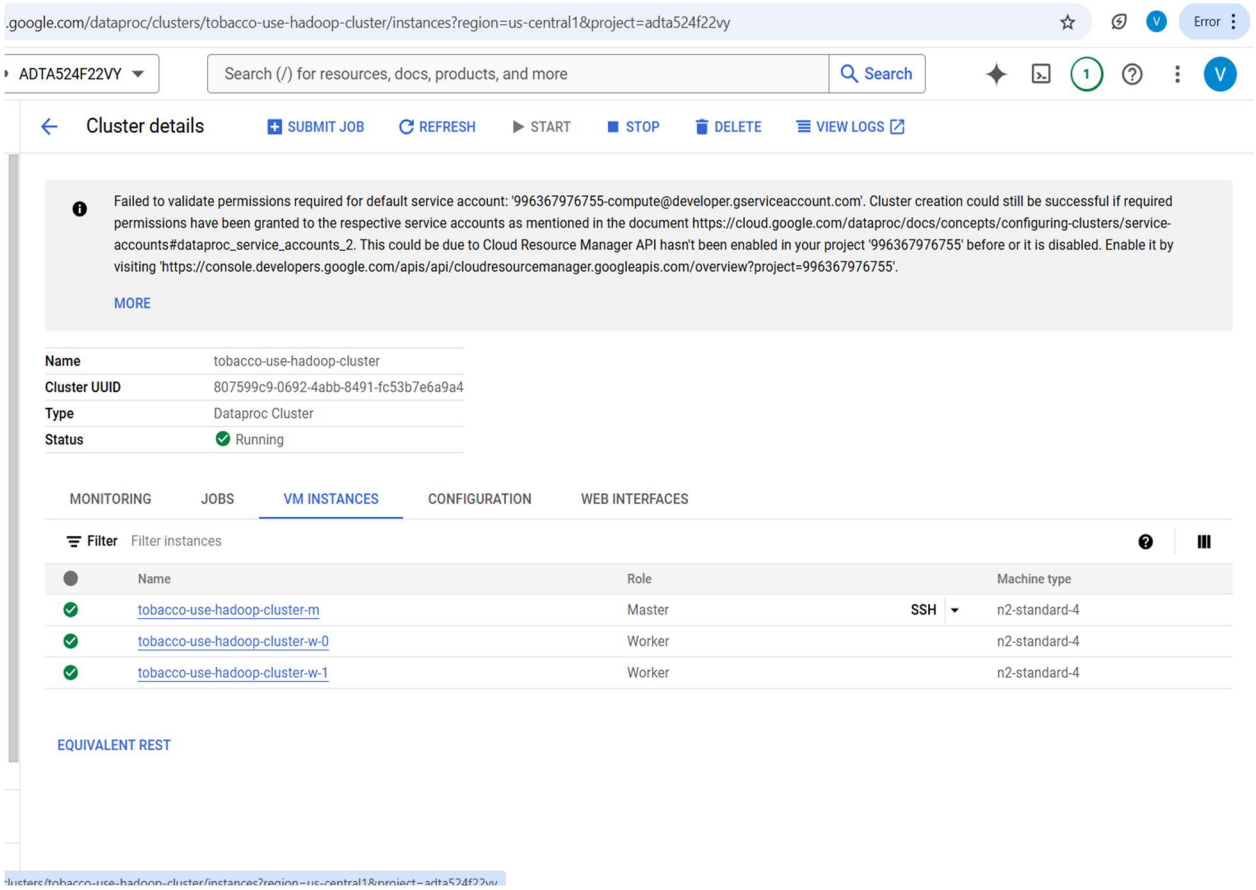
Step 3: Setting up the Hadoop ecosystem

The main aim of using the Hadoop filesystem was to manage and analyze large dataset effectively. The name of the cluster is **tobacco-use-Hadoop-cluster.**

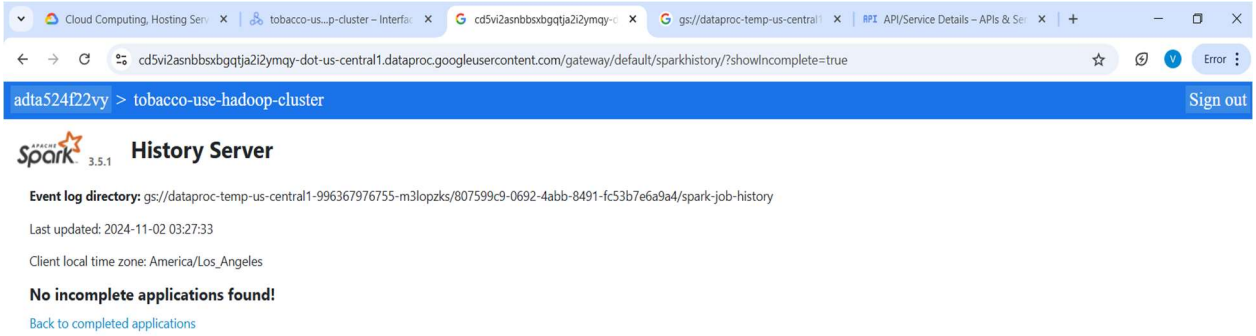
Below is a screenshot of node and worker configuration.



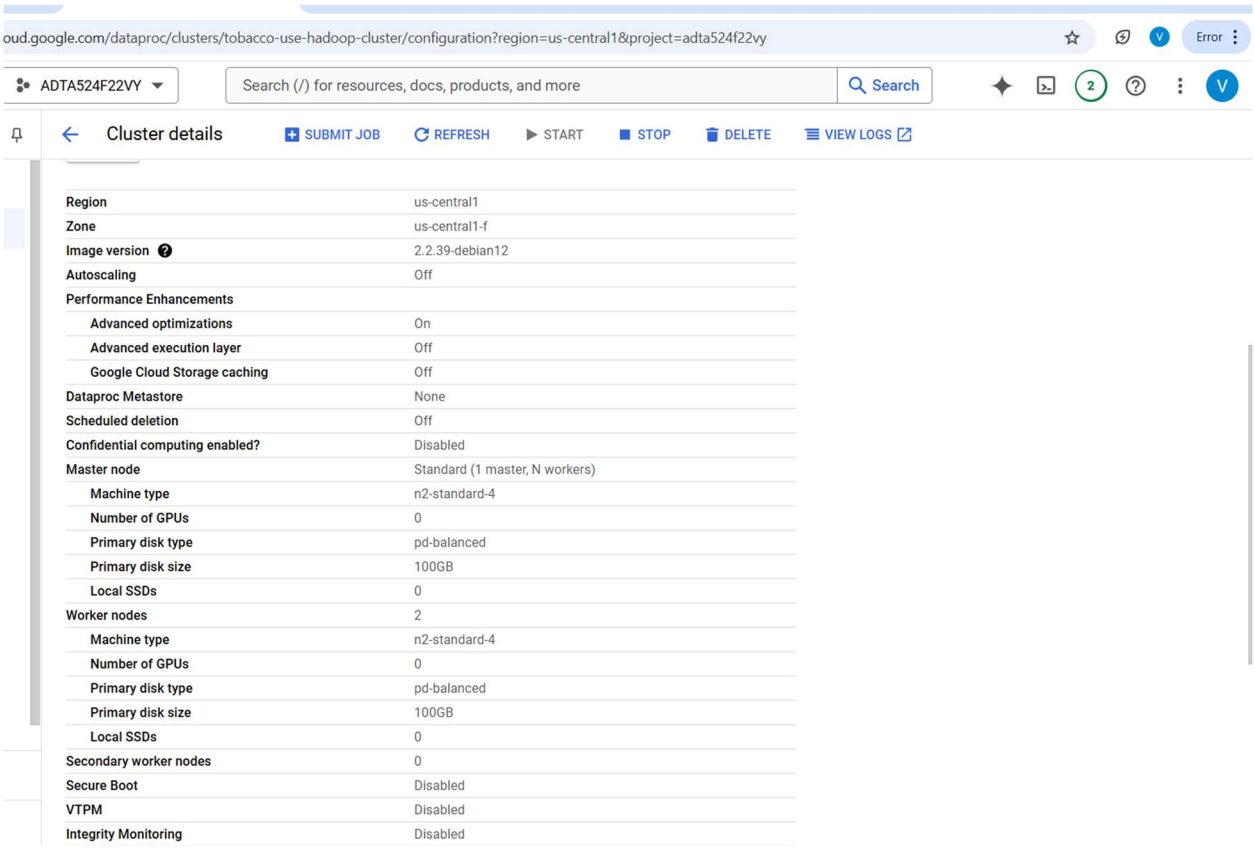
This is the final cluster created with all hive and spark services enabled by default in dataproc for further data processing.



We used the Spark History Server which is a web interface that monitors and reviews Spark jobs status and performance. Using this tool allows us to analyze the details of job executions, read error messages, tune future jobs by seeing resource usage and job running time.



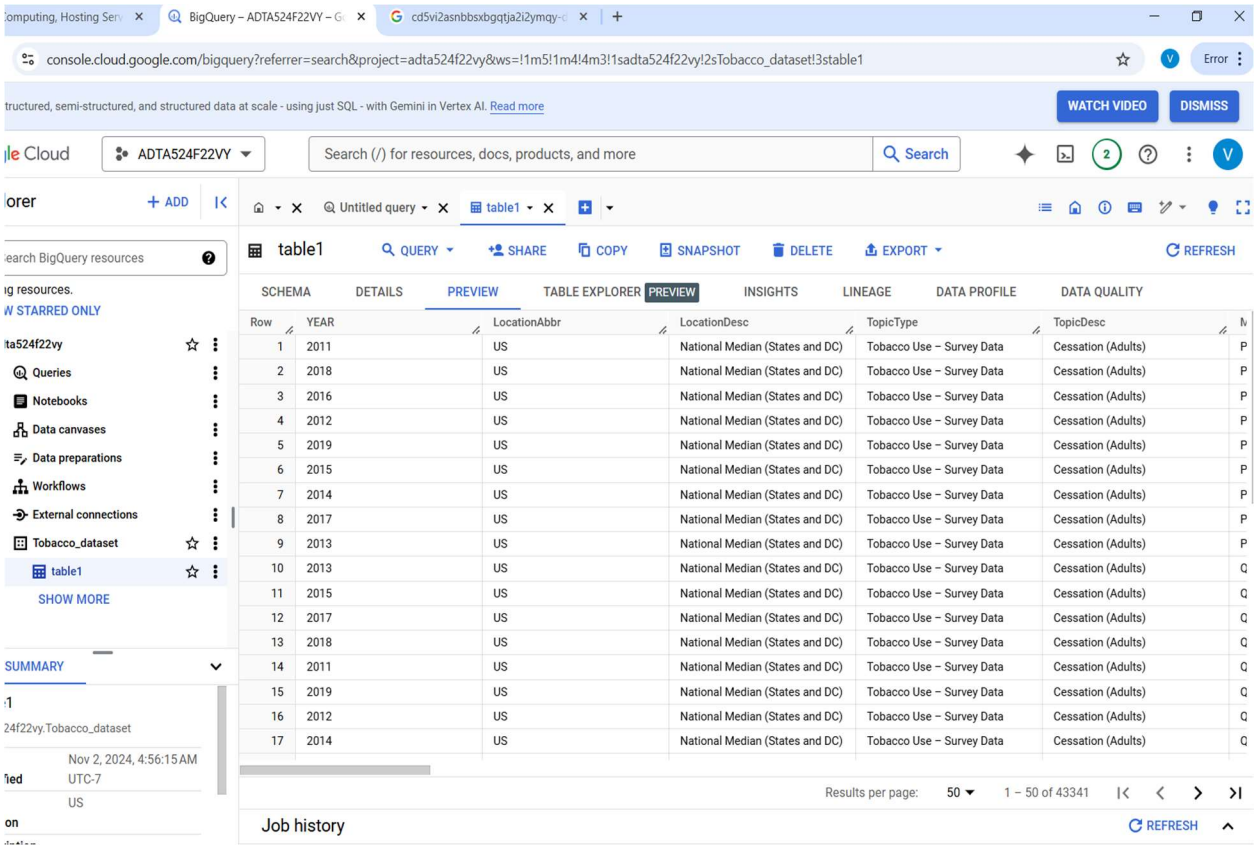
This was the overall configuration of the Hadoop ecosystem



Step 4: Creating dataset and table in Big query for static dataset

In Big query we created a dataset and table where data was ingested for further preprocessing and analysis.

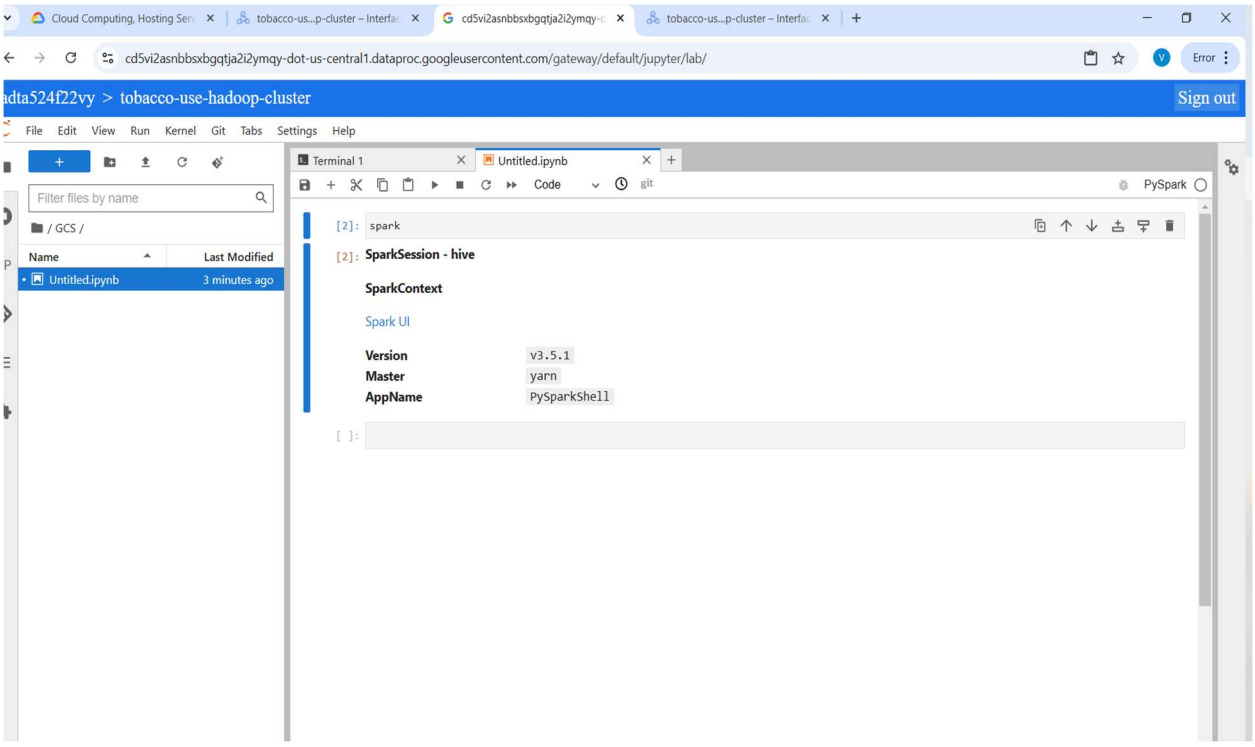
This the table preview for the Tobacco dataset.



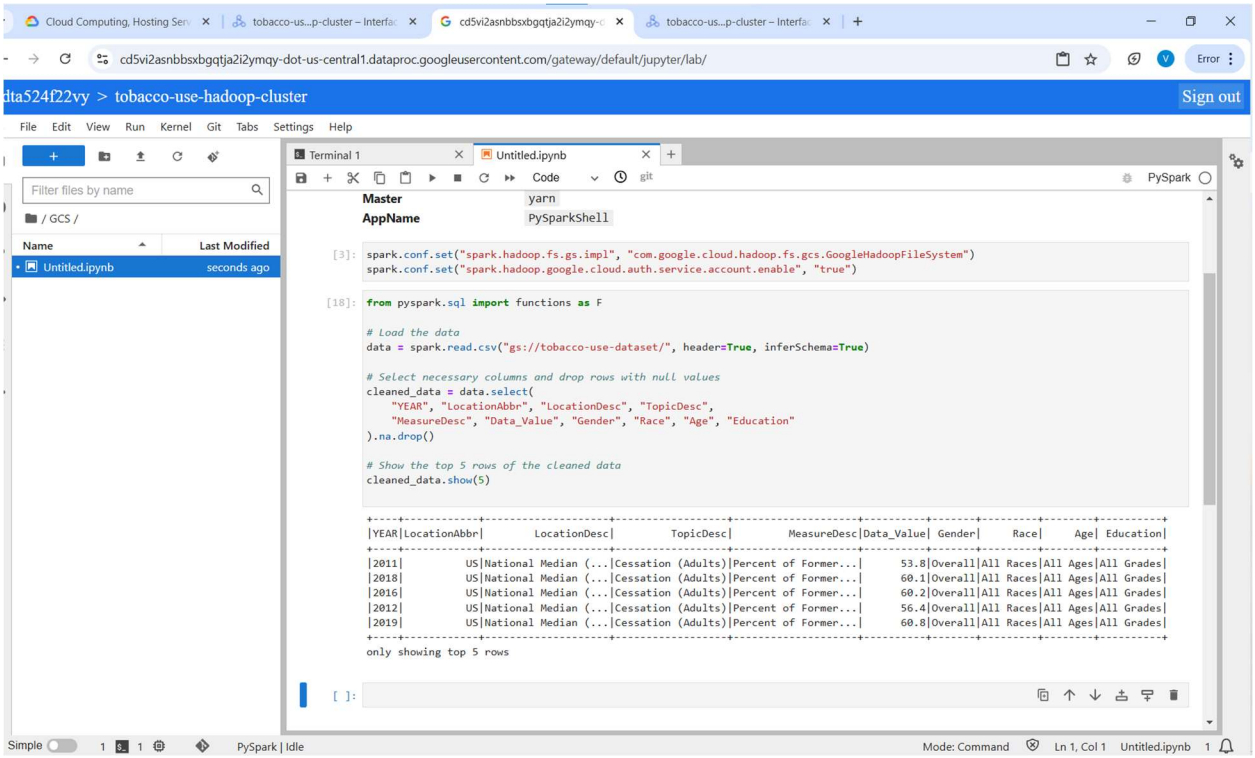
Row	YEAR	LocationAbbr	LocationDesc	TopicType	TopicDesc
1	2011	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
2	2018	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
3	2016	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
4	2012	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
5	2019	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
6	2015	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
7	2014	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
8	2017	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
9	2013	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
10	2013	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
11	2015	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
12	2017	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
13	2018	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
14	2011	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
15	2019	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
16	2012	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)
17	2014	US	National Median (States and DC)	Tobacco Use – Survey Data	Cessation (Adults)

The next step was to export this data to GCS where data was stored in the bucket .

Then because we configured the Hadoop to use the same bucket ,it is easier to do further preprocessing. Data is loaded into pyspark session for further analysis.

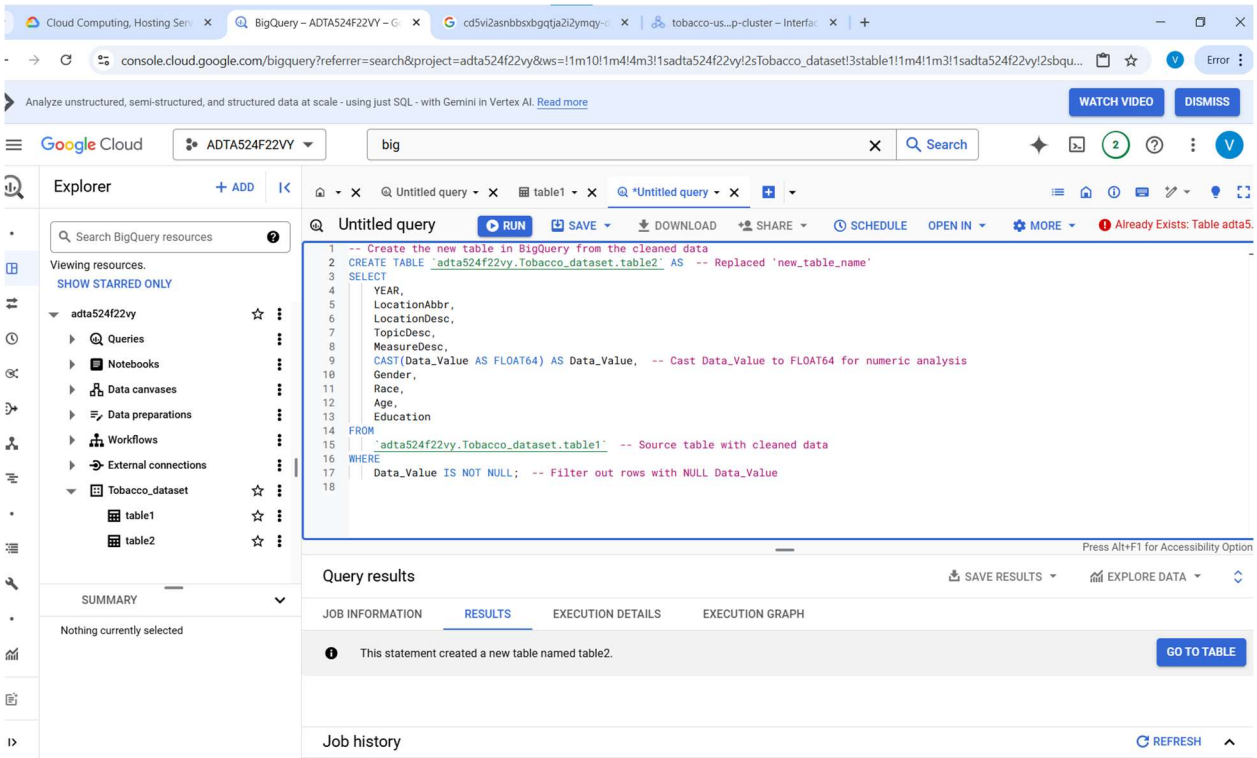


We used spark in data cleaning and filtering all the relevant columns using this code.

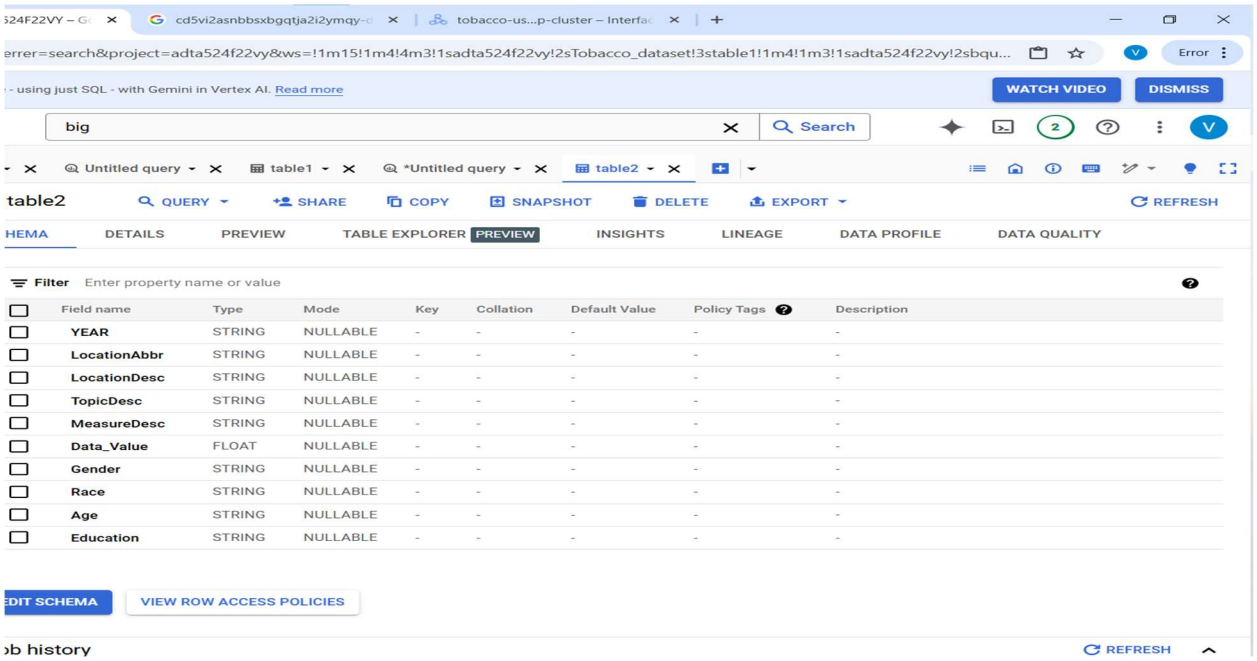


Step 5: advanced data preprocessing and analysis using Big query

The first process was to filter out all unnecessary columns and remove all null values. Then we created a new table in bigquery to store the new cleaned data for further analysis.

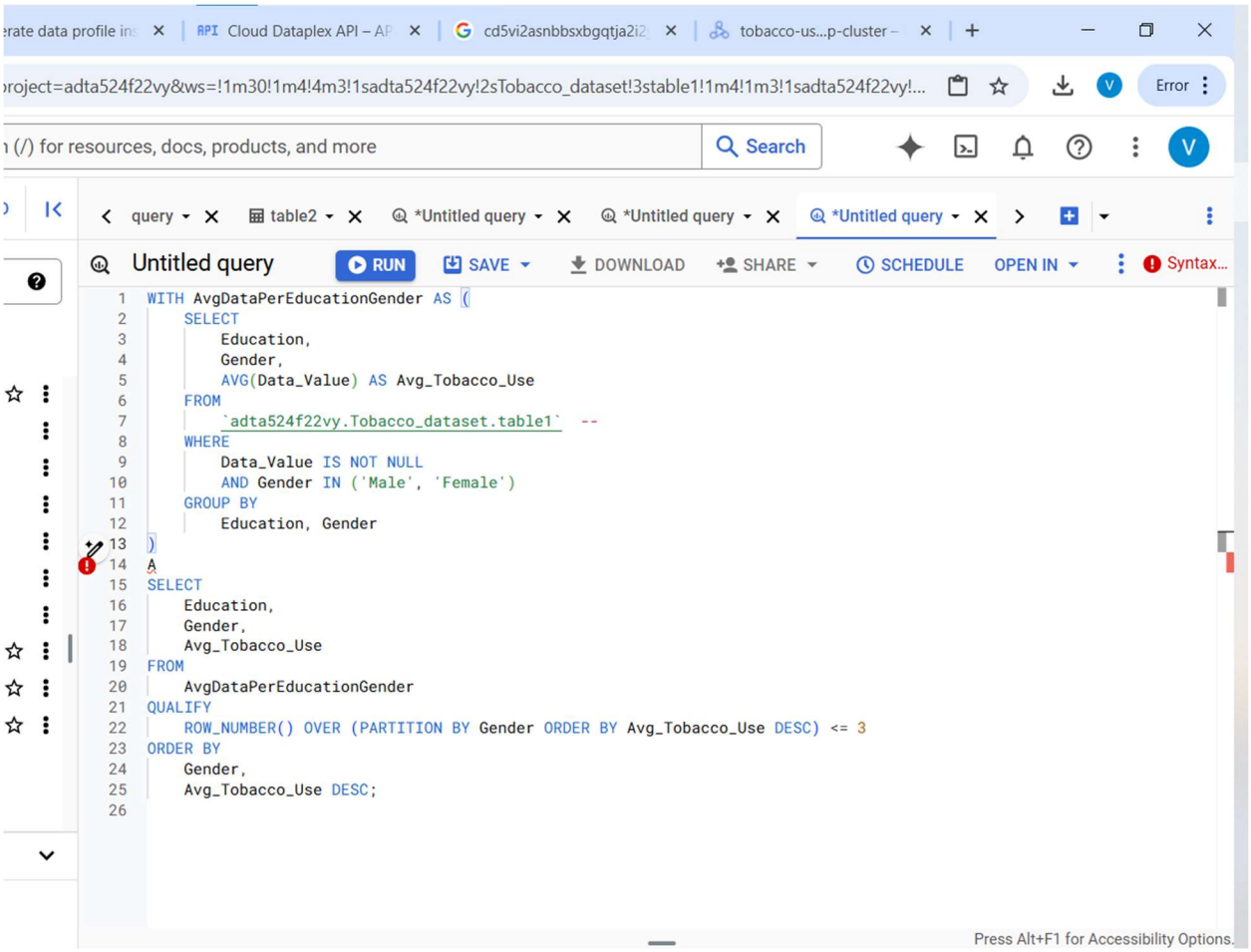


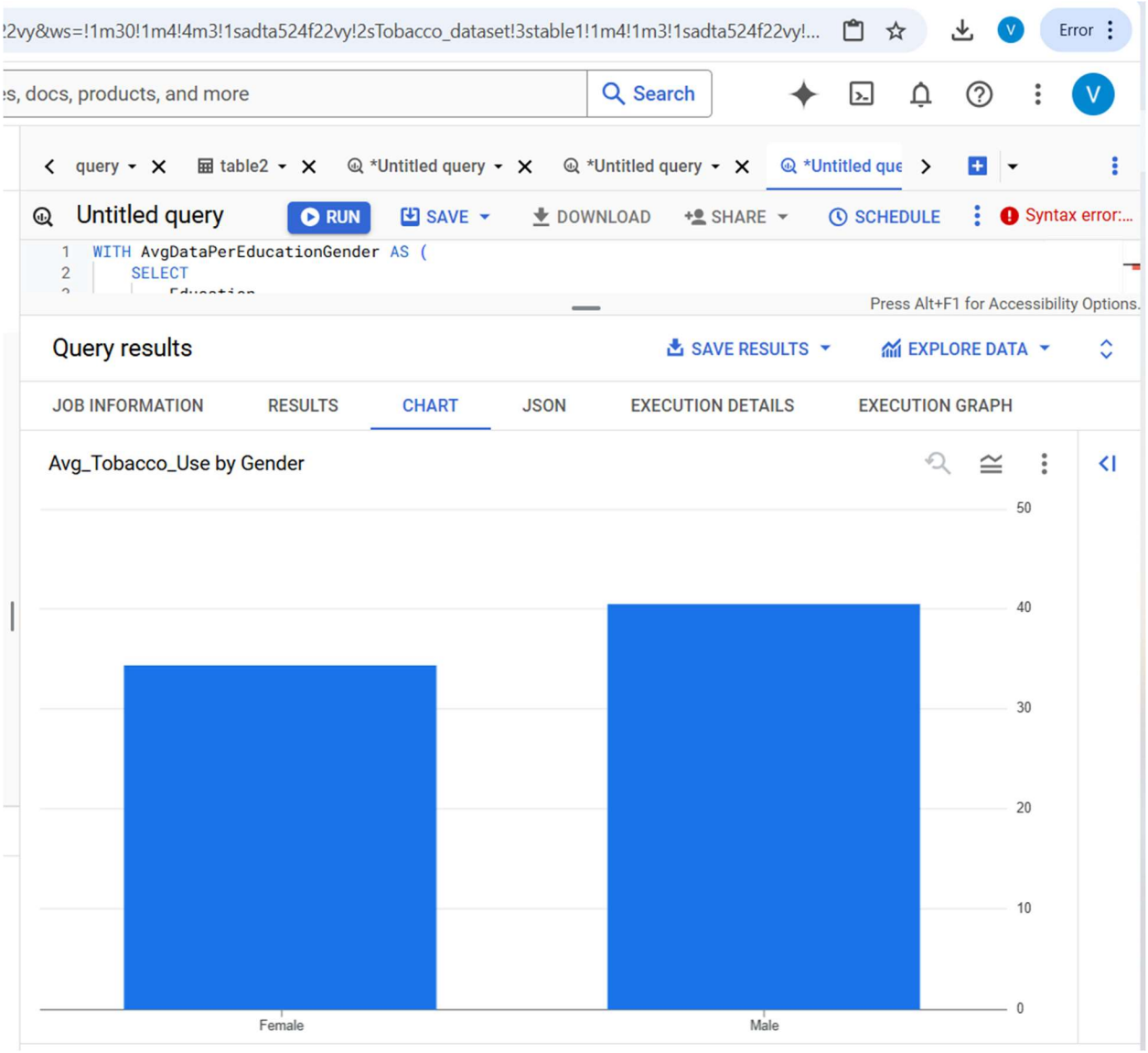
This is the new schema for the cleaned table



Step 7: Data analysis

This query finds out top 3 education level which has highest average of tobacco usage for each of gender (Male, Female). The query calculates the average Data Value (amount of tobacco used), groups by Education and Gender, to show how much tobacco use trends are affecting the population that have had less education, as well as the gender population that has also been put at risk by tobacco.

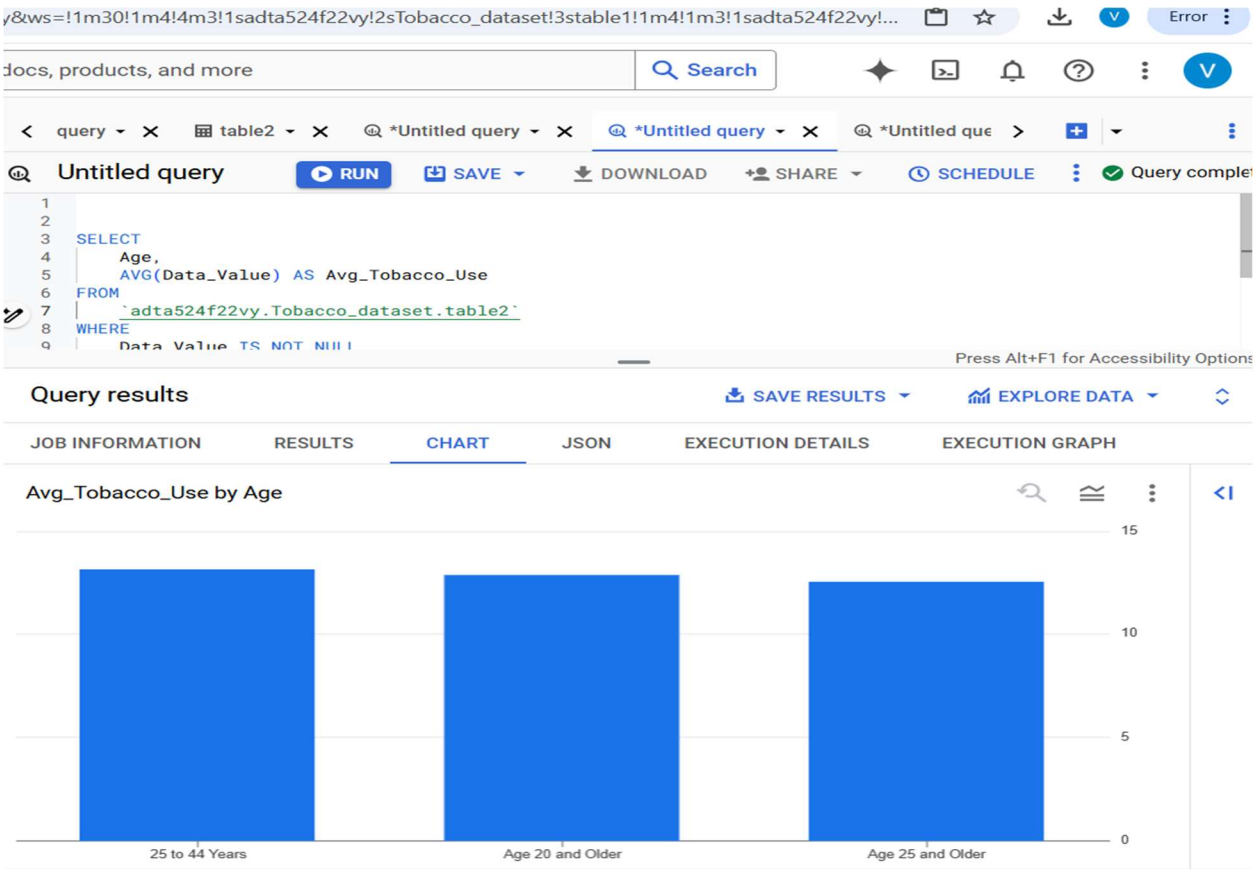




Results show that at all levels of education, males have consistently higher rates of tobacco use than females, suggesting gender differential tobacco use patterns and potential for intervention.

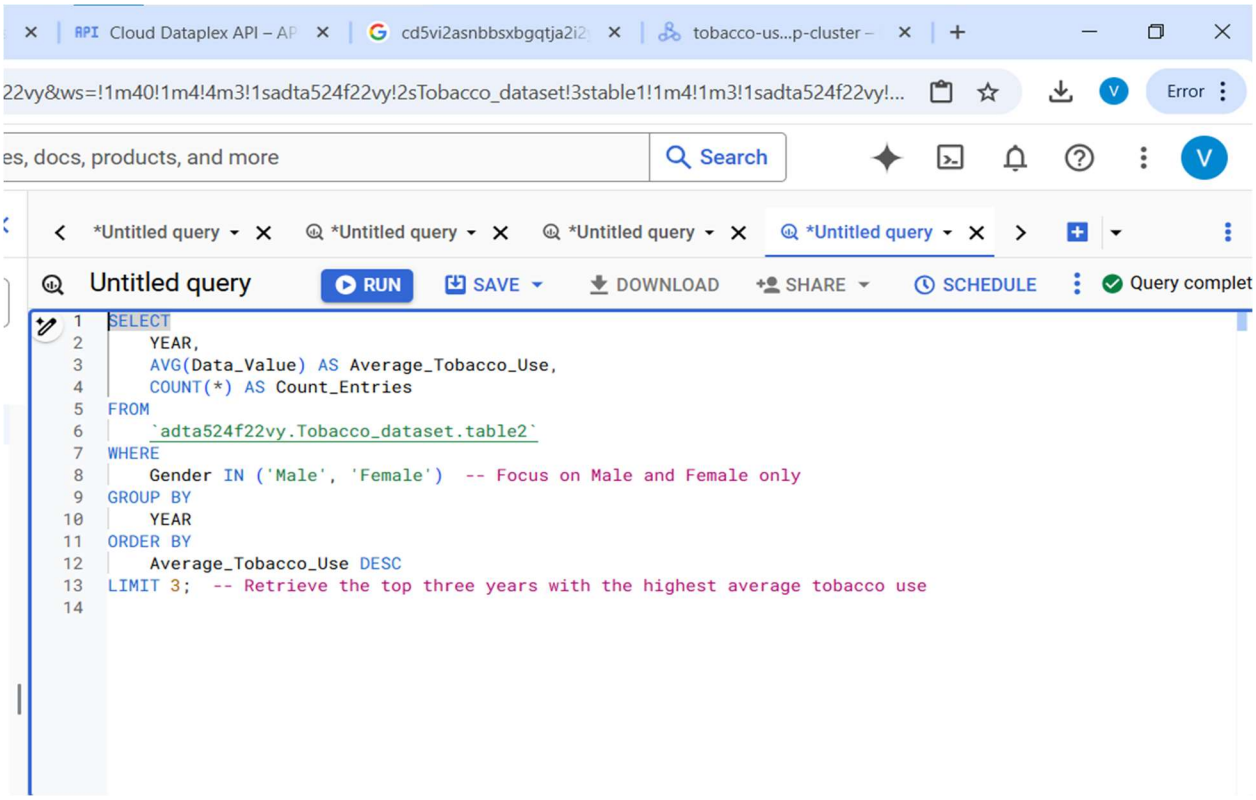
The next analysis was to determine the highest the top three with highest tobacco use

```
SELECT
    Age,
    AVG(Data_Value) AS Avg_Tobacco_Use
FROM
    `adta524f22vy.Tobacco_dataset.table2`
WHERE
    Data_Value IS NOT NULL
    AND Age NOT IN ('All Ages', 'Unknown')  -- Exclude non-specific age categories
GROUP BY
    Age
ORDER BY
    Avg_Tobacco_Use DESC
LIMIT 3;
```



Trend analysis of tobacco usage among different age groups reveals that adults in the age group bracket of 25 to 44 years consume tobacco at the highest average rates. This is especially worrying because it implies that young persons are more prone to get nicotine addicted, thereby becoming a big cause of quite severe health risks. In this group, after this one, the broader categories of those 20 and older and 25 and older, also also have elevated usage rates.

The other insight was the top three years with the highest number of tobacco usage



The relatively stable average tobacco use over these three years suggests a persistent trend of consumption either of an underlying social, economic or policy factor influence on tobacco use. We could further explore these years and see if that's enlightening to us in terms of public health initiatives or legislation affecting how people smoke.

Conclusion

Tobacco use data, which includes all the trends and patterns characteristics across various demographic and year groups, is analyzed in a comprehensive way. Thus, the findings indicate that tobacco consumption continues to be a persistent public health 'concern' at all levels of education, for both genders and in different age groups. High usage rates have been shown consistently in males relative to females implying a potential demographic for health interventions.

To further investigate what may be causing these trends, the data suggests that some years, in 2013, saw peaks of average tobacco consumption. The stable averages over time indicated that tobacco use is both a persistent problem and one that needs continued attention of public health through strategic efforts.

The data provide insights into variations in tobacco use, by race and by age, that can help shape tailored prevention and cessation programs to reduce tobacco related health risk. It might also help guide the timing of public health campaigns based on tobacco consumption patterns by season.

References

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- https://www.cdc.gov/brfss/index.html
- https://healthdata.gov/browse?q=Behavioral+Risk+Factor+Surveillance+System&sortBy=relevance&page=1&pageSize=20